

ARE100B Week 5 Discussion problems

Question 1

J&J and Anova pharmaceutical companies are about to enter a new drug market.

The inverse demand for this new drug is: $P = 250 - 2Q$.

The cost curves for both firms are:

- For J&J: $TC_J = 25q_J$
- For Anova: $TC_A = 35q_A$

Question 1 (a): If both J&J enter and compete in a Cournot game (choosing quantities), find the profit for each firm.

Firm i 's profit:

$$\pi_i = (250 - 2(q_J + q_A))q_i - c_i q_i, \quad c_J = 25, \quad c_A = 35$$

J&J's profit:

$$\pi_J = (250 - 2(q_J + q_A))q_J - 25q_J$$

Finding J&J's best response function:

$$\frac{\partial \pi_J}{\partial q_J} = 225 - 4q_J - 2q_A = 0 \Rightarrow q_J = BR_J = 56.25 - 0.5q_A$$

Anova's profit:

$$\pi_A = (250 - 2(q_J + q_A))q_A - 35q_A$$

Finding Anova's best response function:

$$\frac{\partial \pi_A}{\partial q_A} = 215 - 2q_J - 4q_A = 0 \Rightarrow q_A = BR_A = 53.75 - 0.5q_J$$

Solve simultaneously:

$$q_J = 39.17, \quad q_A = 34.17, \quad Q = 73.33$$

Price:

$$P = 250 - 2(73.33) = 103.33$$

Profits:

$$\pi_J = (103.33 - 25)(39.17) = 3068.06, \quad \pi_A = (103.33 - 35)(34.17) = 2334.72$$

J&J's profit is:

$$\pi_J = 103.33 \times 39.17 - 25 \times 39.17 = 3,068.06$$

Anova's profit is:

$$\pi_A = 103.33 \times 34.17 - 35 \times 34.17 = 2,334.72$$

Question 1 (b): Instead, executives for J&J meet in a smoke-filled room and agree not to compete with each other. Owing to Anova's higher costs, they agree that Anova will not enter the market, and J&J will share 50% of its profit with Anova as a reward for not entering. Find the cartel output and profit for each firm.

Monopoly condition:

$$MR = 250 - 4Q = 25 \Rightarrow Q^M = 56.25$$

Price:

$$P^M = 250 - 2(56.25) = 137.5$$

J&J monopoly profit:

$$\pi_J^M = (137.5 - 25)(56.25) = 6,328.13$$

Cartel agreement: J&J shares 50

$$\pi_J^{cartel} = \pi_A^{cartel} = 3,164.06$$

Question 1 (c): Are the cartel profits high enough to incentive J&J and Anova to collude?

We want to compare the cartel profit for each firm against that they would earn in a Cournot game. If the cartel profit is larger for each firm, they would have the incentive to collude.

$$\pi_J^{Cartel} = 3,164.06 > \pi_J^{Cournot} = 3,068.06, \quad \pi_A^{Cartel} = 3,164.06 > \pi_A^{Cournot} = 2,334.72$$

Thus both firms prefer cartel outcomes so they would collude.

Question 1 (d): If Anova observes J&J following the cartel agreement, what would its cheating profit and output be? Does Anova have the incentive to cheat the cartel?

If J&J's output is the agreed cartel output of 56.25, then Anova's best response for output is:

$$\begin{aligned} q_A = BR_A &= 53.75 - 0.5q_J \\ &= 53.75 - 0.5 \times 56.25 \\ &= 25.625 \end{aligned}$$

So the price would be:

$$P = 250 - 2 \times (56.25 + 25.62) = 86.25$$

So Anova's cheating profit would be:

$$\begin{aligned} \pi_A^{Cheating} &= 86.25 \times 25.625 - 35 \times 25.625 \\ &= 1,313.28 \end{aligned}$$

But this is less than the profit it would make from not entering and getting the side payment from J&J! So Anova does not have the incentive to cheat and the cartel would remain.

Extra practice questions

Question A

Greg and Will run concrete businesses in Davis. Normally they compete in Cournot games (choosing quantities).

The inverse demand for concrete in Davis is $P = 1,250 - 5Q$.

Greg and Will's costs are:

$$TC_G = 27q_G$$

$$TC_W = 27q_W$$

Question A (a): Find the Cournot outputs and profits for Greg and Will's concrete businesses.

Greg's profit function is:

$$\pi_G = (1,250 - 5(q_G + q_W)) \times q_G - 27q_G$$

Greg's best response function is:

$$\begin{aligned}\frac{d\pi_G}{dq_G} &= 1,250 - 27 - 5q_W - 10q_G = 0 \\ q_G &= \frac{1,223 - 5q_W}{10}\end{aligned}$$

Because Will has the same costs and Greg and Will's products are identical, Will's best response function is:

$$q_W = \frac{1,223 - 5q_G}{10}$$

Finding the Nash Equilibrium outputs is:

$$\begin{aligned}q_G &= \frac{1,223 - 5 \frac{1,223 - 5q_G}{10}}{10} \\ q_G &= \frac{1,223 - 611.5 + 2.5q_G}{10} \\ 0.75q_G &= 61.15 \\ q_G &= 81.53\end{aligned}$$

And Will's output will be the same at $q_W = 81.53$.

The market price will be:

$$P = 1,250 - 5 \times (81.53 + 81.53) = 434.7$$

So the Cournot profits will be:

$$\pi_G = \pi_W = 81.53 \times 434.7 - 27 \times 81.53 = \$33,239.78$$

Question A (b): While taking a walk together in the Arboretum, Greg and Will discuss whether they could both make more money from colluding and splitting their combined outputs and profits 50% each. Is this true? Find the collusion/cartel profit and output.

We have that the marginal costs of the cartel is: $MC = 27$.

The cartel's revenue curve is:

$$\text{Revenue} = (1,250 - Q) \times Q$$

So marginal revenue is:

$$MR = 1,250 - 10Q$$

The cartel will output at the monopoly level where $MR = MC$:

$$27 = 1,250 - 10Q$$

$$Q = 122.30$$

Greg and Will split this 50-50 so each output is:

$$q_G = q_W = 61.15$$

The market price will be:

$$\begin{aligned} P &= 1,250 - 5 \times 122.30 \\ &= 638.50 \end{aligned}$$

The cartel profit will be:

$$\begin{aligned} \pi^{\text{Cartel}} &= 638.50 \times 61.15 - 27 \times 61.15 \\ &= \$74,786.45 \end{aligned}$$

And Greg and Will get 50% each:

$$\pi_G^{\text{Cartel}} = \pi_W^{\text{Cartel}} = \$37,393.22$$

Question A (c): If Greg produces concrete at their cartel output, does Will have the incentive to cheat?

If Greg sticks at the cartel agreed output of 61.15, Will's best response is to output:

$$BR_W = \frac{1,223 - 5 \times q_G}{10} = \frac{1,223 - 5 \times 61.15}{10} = 91.725$$

The resulting price would be:

$$P = 1,250 - 5 \times (61.15 + 91.725) = 485.625$$

Will's profit if they cheat is then:

$$\pi_W^{\text{Cheat}} = 485.625 \times 91.725 - 27 \times 91.725 = \$42,067.38$$

Will's profit if they cheat of \$42,067.38 is higher than if they stay in the cartel of \$37,393.22 so Will would have the incentive to cheat!

Question A (d): Greg and Will have been friends for many years. If Will cheats the cartel, Greg will use social pressure to make Will feel bad - which Will thinks is equivalent to losing \$20,000. Will Will still cheat?

Cheating the cartel only makes Will $42,067.38 - 37,393.22 = \$4,674.16$ - not that much more money. If Greg can penalise them \$20,000 of emotional hurt from from cheating, Will will be $\$4,674.16 - \$20,000 = -\$15,325.84$ worse off from cheating. So Will shouldn't cheat!